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Quiz 3b

2/2/2012

1. Use the substitution method to solve the recurrence relation

$$T(n) = T(n-2) - 3$$

assume n is odd

$$T(1) = 2$$

$$T(n-2) = T(n-4) - 3 - 3$$

$$T(n-4) = T(n-6) - 3 - 3 - 3$$

⋮

$$T(1) = 2$$

$$T(3) = 2 - 3$$

$$T(5) = 2 - 3 - 3$$

$$-3 \frac{(n-1)}{2} + 2$$

$$\frac{1}{2} - (3n-1)$$

2. Use the substitution method to solve the recurrence relation

$$T(n) = 2T(n-2) + 1$$

assume n is odd

$$T(1) = 4$$

$$= 4T(n-4) + 2 + 1$$

$$= 8T(n-6) + 4 + 2 + 1$$

$$2^n$$

$$2^{n-1}$$

$$T(3) = 2(4) + 1$$

$$T(5) = 2T(3) + 1$$

3. Use the substitution method to solve the recurrence relation

$$T(n) = 4T(n/2) - 1$$

$$T(1) = T(2) = T(3) = T(4) = 15$$

$$T(n/2) = 4T(n/4) - 4 - 1$$

$$\lg n - 2$$

$$T(n/4) = 4^2 T(n/8) - 16 - 4 - 1$$

$$T(n/8) = 4^3 T(n/16) - 4^3 - 4^2 - 4^1 - 4^0$$

$$\sum_{i=0}^n 4^i$$